Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/07/2019
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	The health risks associated with spaceflight-induced ocular structural and functional damage has become a recent concern for NASA. The goal of the present study was to characterize the effects of space flight and reentry to 1g on the structure and integrity of the retina and blood-retinal barrier (BRB) in the eye. To investigate possible mechanisms, changes in protein expression profiles were examined in mouse ocular tissue after space flight. Ten week old male C57BL/6 mice were launched to the International Space Station (ISS) on Space-X 12 at the Kennedy Space Center (KSC) on August, 2017. After a 35-day mission, mice were returned to Earth alive. Within 38+/-4 hours of splashdown, mice were sacrificed and ocular tissues were collected for analysis. Ground control (GC) and vivarium control mice were maintained on Earth in flight hardware or normal vivarium cages, respectively. Repeated intraocular pressure (IOP) measurements were performed before the flight launch and re-measured before the mice were sacrificed after splashdown. IOP was significantly lower in post-flight measurements compared to that of pre-flight (14.4-19.3 mmHg vs 16.3-20.3 mmHg) ($p<0.05$) for the left eye. Flight group had significant apoptosis in the retina and retinal vascular endothelial cells compared to control groups ($p<0.05$). Immunohistochemical analysis of the retina revealed that an increased expression of aquaporin-4 (AQP-4) in the flight mice compared to controls gave strong indication of disturbance of BRB integrity. There were also a significant increase in the expression of platelet endothelial cell adhesion molecule-1 (PECAM-1) and a decrease in the expression of the BRB-related tight junction protein, Zonula occludens-1 (ZO-1). Proteomic analysis showed that many key proteins and pathways responsible for cell death, cell cycle, immune response, mitochondrial function, and metabolic stress were significantly altered in the flight mice compared to ground control animals. These data indicate a complex cellular response that may alter retina structure and BRB integrity following long-term space flight. Mice ocular tissues were also collected for genomics analysis. RNA sequencing detected 600 differentially expressed genes (DEGs) in space flight retinas, which were enriched for genes related to visual perception, the phototransduction pathway, and numerous retina and photoreceptor phenotype categories. Twelve DEGs were associated with retinitis pigmentosa, characterized by dystrophy of the photoreceptor layer rods and cones. Differentially expressed transcription factors indicated changes in chromatin structure, offering clues to the observed phenotypic changes. Immunofluorescence assays showed degradation of cone photoreceptors and increased retinal oxidative stress. Total retinal thickness, retinal pigment epithelium and choroid layer thickness, were significantly lower after space flight. These results indicate that retinal performance may decrease over extended periods of space flight and cause visual impairment. A hematology analyzer was used to quantify white blood cell (WBC), red blood cell (RBC), and platelet (PLT) parameters in space flight mice compared to ground controls. While there was no significant difference in overall WBC counts in flight compared to ground control mice, there was a significant decrease ($P<0.05$) in lymphocyte (LYM) counts as well as a significant increase ($P<0.05$) in granulocytes (GRA) in flight mice. Furthermore, there was a significant shift ($P<0.05$) in WBC populations away from LYM and towards monocytes (MON) and GRA, suggesting that space flight alters immune function. Additionally, RBC mean corpuscular volume (MCV) was significantly decreased ($P<0.05$) after space flight, which also significantly increased ($P<0.05$) RBC distribution width (RDW). These disruptions in erythrocyte populations may indicate space flight-induced anemia. PLT counts were also significantly lower ($P<0.001$) compared to ground controls, while mean PLT volume (MPV) significantly increased ($P<0.05$). This could also be an indication of a change in cellular function due to space flight. These results indicate space flight dramatically alters circulating cell populations, which are sustained for at least 36 hours after splashdown.
Bibliography Type:	Description: (Last Updated: 07/10/2026)

Articles in Peer-reviewed Journals	Mao XW, Nishiyama NC, Byrum SD, Stanbouly S, Jones T, Drew A, Sridharan V, Boerma M, Tackett AJ, Zawieja D, Willey JS, Delp M, Pecaut MJ. "Characterization of mouse ocular response to a 35-day spaceflight mission: Evidence of blood-retinal barrier disruption and ocular adaptations." Sci Rep. 2019 Jun 3;9(1):8215. https://doi.org/10.1038/s41598-019-44696-0 ; PubMed PMID: 31160660; PubMed Central PMCID: PMC6547757 , Jun-2019
Articles in Peer-reviewed Journals	Overbey EG, da Silveira WA, Stanbouly S, Nishiyama NC, Roque-Torres GD, Pecaut MJ, Zawieja DC, Wang C, Willey JS, Delp MD, Hardiman G, Mao XW. "Spaceflight influences gene expression, photoreceptor integrity, and oxidative stress-related damage in the murine retina." Sci Rep. 2019 Sep 16;9(1):13304. https://doi.org/10.1038/s41598-019-49453-x ; PubMed PMID: 31527661; PubMed Central PMCID: PMC6746706 , Sep-2019